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The Effects of Policy Interventions to Limit Illegal Money Lending

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1 Big picture

- **Question.** What policies are effective at limiting illegal money lending (IML), and through which borrower and lender incentives do they work?
- **Setting.** The paper studies loan sharks in Singapore, using detailed survey data on borrowers and loans in a market organized by syndicates.
- **Data advantage.** The authors observe both desired loan amounts and granted loan amounts, lender identity, interest rates, repayment history, missed payments, and harassment outcomes. This is unusually rich for an illegal credit market.
- **Central market feature.** Lenders charge extremely high interest rates, use harassment to enforce repayment, and use a loan-reset structure that can trap borrowers in repeated payment cycles.
- **Main empirical shock.** During the sample period, law enforcement increased the cost of harassment. The cartel response was to raise the interest rate from 20% to 35%.
- **Model contribution.** The paper estimates a structural model with borrower loan demand, borrower lender choice, lender loan-size choice, lender harassment choice, borrower effort, borrower present bias, and lender estimates of repayment ability.
- **Main crackdown result.** The enforcement crackdown reduced total lender profits from S\$3.08 million to S\$1.00 million, a 67.7% decline, and reduced loan volume by 48.6%.
- **But borrower welfare also falls.** Despite lower lender profits, borrower surplus falls by 12.4% because borrowers receive smaller loans, pay higher interest rates, miss more payments, and are harassed more often.
- **Mechanism.** The crackdown raises the unit cost of harassment. Lenders respond by reducing harassment probabilities and loan sizes, while the cartel raises the interest rate. Higher rates worsen repayment and effort incentives.
- **Targeting borrowers.** Removing the lowest repayment-ability borrowers is not very effective because they are costly to serve. Removing middle-repayment-ability borrowers hurts lenders most because they generate both loan volume and penalties.
- **Indirect interventions.** Policies reducing gambling, drug use, present bias, or heavy discounting lower lender profitability and generally improve borrower surplus.
- **Economic takeaway.** Illegal lending is not just high-price lending. It is a market where monitoring and enforcement are criminalized. Policies can attack lender profits either by increasing enforcement costs or by reducing borrower demand and present-biased repayment behavior.

2 Data and measurement

2.1 Market and institutional setting

- Illegal money lending combines high interest rates, illegal enforcement, and attempts to keep vulnerable borrowers in debt.
- Borrowers in the sample are rejected by formal credit markets and turn to loan sharks as lenders of last resort.
- The market is coordinated by syndicates that standardize key contract terms, including the interest rate and reset structure.
- Individual lenders choose whether to lend, how much to lend, and how likely they are to harass after missed payments.

Interpretation. The model treats interest rates as cartel-level choices and loan sizes/harassment as lender-level choices. This distinction matters because the policy shock changes both lender costs and the cartel interest rate.

2.2 Loan contract and repayment structure

- A typical loan is repaid through weekly payments over six successful payment weeks.
- If the borrower misses a payment, the loan resets and the borrower must restart the sequence of successful payments.
- If the borrower misses consecutive payments, harassment becomes more severe.
- The first payment is withheld at origination, so the lender disburses only part of the face amount while charging interest on the full amount.

Interpretation. The loan reset structure is central. Lender profits come not only from stated interest, but also from penalties and repeated extensions caused by missed payments.

2.3 Survey and panel data

- The full dataset contains 11,032 loans taken out by 1,090 borrowers over 2009–2016.
- The estimation subsample in the summary tables contains 8,836 loans and 1,057 borrowers.
- For each loan, the paper observes desired loan size, granted loan size, lender identity, interest rate, repayment duration, missed payments, and whether harassment occurred.
- Borrower characteristics include demographics, addictions, criminal background, gang membership, past loan history, risk aversion, and discounting.

Interpretation. The data allow the authors to separately model demand for credit, supply of credit, lender harassment, and borrower repayment behavior.

2.4 Key variables

- **Loan size:** amount granted by the lender.
- **Desired loan size:** amount requested by the borrower.

- **Interest rate:** cartel-set nominal loan interest rate.
- **Time to repay:** number of weeks until the loan is completed.
- **Missed payments:** number of weeks in which the borrower fails to make the required payment.
- **Harassment:** whether the lender used harassment at least once during the loan.
- **Past relationship:** number of past loans and missed payments with the same lender.

Interpretation. These variables are mapped directly to the model: loan demand, repayment ability, lender information, harassment costs, and borrower disutility.

2.5 The crackdown

- The major policy intervention is a crackdown on lenders during the sample period.
- The crackdown raises the cost of harassment and causes some lenders to exit.
- After the crackdown, the cartel interest rate rises from 20% to 35%.
- The paper uses the model to compare observed outcomes after the crackdown with a counterfactual no-crackdown economy.

Interpretation. The policy is not a simple price regulation. It changes enforcement technology, loan sizes, interest rates, lender profitability, and borrower welfare simultaneously.

3 Model and empirical specifications

3.1 Borrower loan demand

The borrower receives a borrowing need and chooses a desired loan amount

$$L_{it}^* = \exp(\alpha_i r_t + \theta_i^\alpha + \nu_{it}), \quad (1)$$

where r_t is the interest rate, α_i is borrower i 's interest-rate sensitivity, θ_i^α is a borrower demand fixed effect, and ν_{it} is a demand shock. The price sensitivity is modeled as

$$\alpha_i = \boldsymbol{\theta}_r^\alpha \cdot \mathbf{x}_i^\alpha. \quad (1)$$

Interpretation. Borrower demand is decreasing in the interest rate. Heterogeneity in α_i allows different borrowers to react differently to the cartel's interest-rate increase.

3.2 Borrower consideration set

Borrower i chooses among a consideration set

$$\mathcal{C}_{it} \subseteq \{1, \dots, \mathcal{L}\}, \quad (2)$$

plus the outside option of no loan. The consideration set contains the last three lenders from whom the borrower borrowed and one new lender drawn from the observed borrowing-lending network.

Interpretation. Borrowers do not frictionlessly access every lender. Past relationships and referral networks shape which lenders are realistically available.

3.3 Lender harassment probability

The lender chooses a harshness level h_{ilt} , and the probability of harassment after a missed payment is

$$p_{ilt}^\eta(L_{ilt}, h_{ilt}) = \Phi \left(\sum_{h \in \mathcal{H}} \mathbf{1}\{h_{ilt} = h\} \theta_h^\eta + \theta_{k(\ell)}^\eta + \theta_L^\eta L_{ilt} + \boldsymbol{\theta}_x^\eta \cdot \mathbf{x}_{ilt}^\eta \right), \quad (3)$$

where $k(\ell) \in \{\text{Regular, Harsh, Very Harsh}\}$ is lender type and $\Phi(\cdot)$ is the standard normal cdf.

Interpretation. Harassment is an enforcement technology. It depends on lender type, loan size, borrower characteristics, and past lending history.

3.4 Borrower cash, effort, and repayment probability

The borrower's weekly baseline cash is

$$m_{itw}^0 = \max\{0, m_{it}^0 + \nu_{itw}\}, \quad \nu_{itw} \sim N(0, \sigma_i^2). \quad (4)$$

The borrower can increase cash through costly effort. The cash available for repayment is written as

$$m_{itw}(h_{ilt}) = \max\{0, m_{it}(h_{ilt}) + \nu_{itw}\}. \quad (5)$$

The repayment probability in week w is

$$p_{ilt}^m(L_{ilt}, h_{ilt}) = \Phi \left(\frac{m_{it}(h_{ilt}) - r_t L_{ilt}}{\sigma_i} \right). \quad (8)$$

Interpretation. Higher interest rates lower repayment probability mechanically by raising the weekly payment and behaviorally by reducing effort incentives.

3.5 Lender perceived repayment ability

Before observing all borrower traits, a lender estimates repayment probabilities using the information available from past loan history. The lender's perceived repayment probability is

$$\tilde{p}_{i\ell t}^m(L_{i\ell t}, h_{i\ell t}) = \Phi\left(\frac{\tilde{m}_{i\ell t}(h_{i\ell t}) - r_t L_{i\ell t}}{\tilde{\sigma}_i}\right). \quad (10)$$

Interpretation. The model separates the borrower's true repayment ability from the lender's estimate of it. This matters because lenders screen and learn from past borrowing history.

3.6 Lender expected payoff

At origination, the lender disburses only $(1 - r_t)L_{i\ell t}$, giving payoff

$$\tilde{u}_{i\ell 1}(L_{i\ell t}) = -(1 - r_t)L_{i\ell t}. \quad (11)$$

If a payment is missed, the lender may harass with probability $p_{i\ell t}^\eta$. The expected cost of harassment is

$$\kappa_{\ell t} = c_{\ell t} + p_{\ell t}^e K_{\ell t}, \quad (12)$$

where $c_{\ell t}$ is the direct cost, $p_{\ell t}^e$ is the probability of arrest, and $K_{\ell t}$ is the expected disutility from arrest. The paper parameterizes this cost as

$$\kappa_{\ell t} = (1 + \theta_{k(\ell)}^\kappa)(\theta_0^\kappa + \theta_{post}^\kappa post_t). \quad (14)$$

Interpretation. The crackdown enters as a rise in θ_{post}^κ , the post-crackdown harassment-cost component.

3.7 Borrower lender choice

For each possible lender, borrowers form an expected present value $\bar{V}_{i\ell t}$. Their lender choice probability has a logit form:

$$\Pr\left(V_{i\ell t} > \max_{\ell' \in \{0\} \cup \mathcal{C}_{it} \setminus \{\ell\}} V_{i\ell' t}\right) = \frac{\exp(\bar{V}_{i\ell t})}{\sum_{\ell' \in \{0\} \cup \mathcal{C}_{it}} \exp(\bar{V}_{i\ell' t})}. \quad (38)$$

Interpretation. Borrower choice identifies harassment disutility and effort costs because lenders differ in expected loan size, expected harshness, and past relationship history.

3.8 Policy counterfactuals

The model is used to simulate counterfactuals for:

- no crackdown versus observed crackdown;
- only harassment-cost changes versus only interest-rate changes;
- alternative cartel interest rates;
- removing groups of borrowers based on repayment ability;
- reducing gambling, drug use, present bias, or heavy discounting.

Interpretation. The counterfactuals are the payoff from structural estimation. Reduced-form facts show what happened; the model decomposes why it happened and what alternative policies would do.

4 Identification and empirical design

4.1 What identifies the model parameters

- Harassment events, missed payments, time to repay, and borrower/loan characteristics identify the harassment probability parameters.
- Missed payments and time to repay identify repayment probabilities from both the borrower and lender perspectives.
- Observed loan sizes identify the lender's cost of harassment because lenders trade off expected interest revenue against expected harassment costs.
- Desired loan sizes and the interest-rate increase identify borrower demand and price sensitivity.
- Borrower lender choices identify harassment disutility and effort costs through variation in expected loan sizes, harassment probabilities, and lender histories.

Interpretation. The paper's structural identification relies on observing both sides of the market: what borrowers ask for, what lenders grant, how repayment unfolds, and whether enforcement occurs.

4.2 Why the crackdown is useful

- The crackdown raises the expected cost of harassment by making enforcement riskier for lenders.
- The cartel responds by raising the interest rate from 20% to 35%.
- This gives a sharp source of variation in the price of credit and in the cost of enforcement.
- The model decomposes the combined observed effect into a cost channel and an interest-rate channel.

Interpretation. The same policy shock changes lender costs and borrower prices. The model is needed because the two channels have opposite effects on some outcomes.

4.3 Why structural modeling is needed

- Borrower welfare is not observed directly.
- Lender profits depend on counterfactual repayment paths, harassment costs, loan sizes, and choices of harshness.
- The policy interest is not only what happened in 2014, but what would happen under different enforcement intensities or borrower-targeting policies.
- The model can simulate no-crackdown and alternative-intervention worlds.

Interpretation. Descriptive evidence can show that loan sizes fell and harassment changed. Structural estimates are needed to infer profits, welfare, and alternative policy effects.

4.4 Important assumptions and limits

- The cartel interest rate is taken as given in the baseline model rather than endogenized.
- Borrowers are assumed not to strategically default when they can afford a payment.
- Borrowers are assumed not to save across weeks during a loan.

- The data observe whether harassment occurred in a loan, but not the exact week of harassment.
- The data are survey-based and come from a vulnerable population, although the sampling strategy is designed to reduce selection concerns.

Interpretation. These assumptions are reasonable for the data available, but they should be kept in mind when interpreting welfare and counterfactual results.

5 Tables and figures

This section uses the same *Read* plus *Economic message* structure as the earlier notes. Adjacent tables and figures are grouped to save space and speed up comparison. All visuals are cropped directly from the original paper.

5.1 Tables 1–3: loan and borrower summary statistics

Read:

- Table 1 shows that the average loan size is S\$1,288.56, while the average desired loan size is S\$1,600.55.
- The average interest rate is 22.28%, loans take 13.38 weeks to repay, and borrowers miss 3.85 payments on average.
- Harassment is common: 54% of loans involve harassment at least once.
- Table 2 shows the sharp post-crackdown shift: interest rates rise to about 38.3% in 2015–2016, loan sizes fall sharply, and harassment incidence rises to roughly 89%.
- Table 3 shows that borrowers are financially and socially vulnerable: 90% gamble, 97% drink alcohol, 31% use drugs, 69% frequent sex workers, and 30% were previously gang members.

Economic message. The loan data reveal a market with high rates, repeated missed payments, and frequent harassment. The borrower data explain why standard formal credit is unavailable and why policy interventions might target borrower behavior as well as lenders.

Table 1
Summary statistics of loan-level variables.

	N	Mean	Std. Dev.	Min.	Median	Max.
Loan size (in S\$)	8836	1288.56	983.28	300	1000	5000
Desired loan size (in S\$)	8836	1600.55	1018.65	300	1000	5000
Interest rate (in %)	8836	22.28	6.48	2	20	50
Number of weeks to repay	8836	13.38	5.84	6	12	24
Number of missed payments	8836	3.85	3.91	0	2	23
Number of past loans with lender	8836	4.10	3.43	0	3	19
Worked for lender to repay	8547	0.06	0.24	0	0	1
Harassed at least once in loan	8836	0.54	0.50	0	1	1

The statistics shown are for subsample of data used in estimation.

Table 2
Means of loan-level variables by year.

	2009	2010	2011	2012	2013	2014	2015	2016
Loan size (in S\$)	1488.51	1403.75	1456.85	1537.84	1506.85	972.14	425.72	480.43
Desired loan size (in S\$)	1701.01	1664.55	1742.50	1838.79	1847.72	1378.89	910.54	966.67
Interest rate (in %)	19.33	19.43	19.44	19.48	19.48	30.33	35.31	38.33
Number of weeks to repay	11.94	12.11	12.06	12.00	13.11	15.81	19.21	19.68
Number of missed payments	2.68	3.01	3.18	3.31	4.15	5.34	7.09	6.85
Number of past loans with lender	4.27	3.57	3.51	3.62	3.19	2.18	7.44	4.75
Worked for lender to repay	0.08	0.05	0.04	0.04	0.04	0.11	0.11	0.12
Harassed at least once in loan	0.50	0.46	0.45	0.43	0.48	0.81	0.89	0.88

The statistics shown are for subsample of data used in estimation.

(a) Table 1: loan-level summary statistics.

(b) Table 2: loan-level variables by year.

Summary statistics of borrower characteristics.						
	N	Mean	Std. Dev.	Min.	Median	Max.
Age	1057	37.56	7.64	20	38	63
Post-secondary education	1057	0.19	0.39	0	0	1
Female	1057	0.10	0.30	0	0	1
Married	1057	0.49	0.50	0	0	1
Divorced	1057	0.16	0.36	0	0	1
Has children	1057	0.62	0.49	0	1	1
Malaysian	1057	0.14	0.35	0	0	1
Indian	1057	0.11	0.31	0	0	1
Current gang member	1057	0.14	0.35	0	0	1
Previously gang member	1057	0.30	0.46	0	0	1
Number of previous convictions	1057	0.49	1.11	0	0	6
Gambles	1057	0.90	0.29	0	1	1
Drinks alcohol	1057	0.97	0.18	0	1	1
Uses drugs	1057	0.31	0.46	0	0	1
Frequent sex workers	1057	0.69	0.46	0	1	1
Frequently treats friends	1057	0.09	0.29	0	0	1

The statistics shown are for subsample of data used in estimation.

(c) Table 3: borrower characteristics.

5.2 Tables 4–6: identification and structural estimates

Read:

- Table 4 maps each parameter block to data variation: harassment events, missed payments, time to repay, loan size choices, demanded loan sizes, and lender choices.

- Table 5 estimates harassment costs and behavioral parameters. The estimated baseline harassment cost is S\$437 before the crackdown and rises by about S\$1,050 after the crackdown.
- Harsh and very harsh lenders have lower direct harassment costs, helping explain why they use more severe enforcement more often.
- Table 6 shows that loan demand is strongly decreasing in the interest rate. The paper reports a price elasticity of loan demand around -0.817 .
- Gamblers are less price sensitive, consistent with addiction-related demand for liquidity.

Economic message. The estimated model says lender harassment is costly but profitable, and borrower demand is price-sensitive enough that a rate increase strongly lowers loan volume.

Table 4

Summary of the variation in the data that identify our parameters.

	Parameters determine:	Identified through:
θ^l	Lender's probability of severe harassment after a missed payment.	Observed harassment events the data, given the observed number of missed payments, time to repay, and borrower and loan characteristics.
θ^l, θ^r	Borrower repayment probabilities from lender's perspective.	Observed number of missed payments and time to repay given observed borrower and loan characteristics.
θ^r	Lender's cost of harassment.	Observed loan sizes, where lenders trade off the expected revenue from a loan of a given size with the expected cost of harassment at that loan size (determined by the harassment cost, repayment probability and harassment probability at that loan size).
θ^r	Borrower loan demand parameters.	Observed borrower loan demand sizes, where the change in the interest rate from the enforcement event (supply-side shock) identifies the price elasticity.
θ^r, θ^l	Borrower repayment probabilities from borrower's perspective	Observed number of missed payments and time to repay given observed borrower and loan characteristics.
θ^l, θ^r	Borrower harassment disutility and effort cost	Observed choices of lenders by borrowers, where borrowers trade off the expected loan size with the expected disutility of harassment (θ^l) and expected cost of effort (θ^r), which vary by lender type, harassment probabilities, and past lending history.

(a) Table 4: identification of parameters.

Table 6

Borrower demand estimates.

	Log Loan Asked	
Interest rate	-5.818	(0.906)
Interest rate $\times$ Age	-0.039	(0.015)
Interest rate $\times$ Post-secondary education	0.075	(0.242)
Interest rate $\times$ Female	2.059	(0.326)
Interest rate $\times$ Married (rel. to single)	0.112	(0.426)
Interest rate $\times$ Divorced (rel. to single)	0.047	(0.449)
Interest rate $\times$ Has children	-0.205	(0.428)
Interest rate $\times$ Malaysian (rel. to Singaporean Chinese)	0.766	(0.285)
Interest rate $\times$ Indian (rel. to Singaporean Chinese)	0.740	(0.321)
Interest rate $\times$ Drinks alcohol	0.696	(0.512)
Interest rate $\times$ Uses drugs	0.345	(0.213)
Interest rate $\times$ Frequents sex workers	0.028	(0.237)
Interest rate $\times$ Gamblers	2.907	(0.382)
Borrower fixed effects	Yes	
Number of observations	10269	

Robust standard errors in parentheses clustered at the borrower level.

(b) Table 6: borrower demand estimates.

Table 5

Parameter estimation results.

	Cash for repayments (borrower)		Cash for repayments (lender)		Lender harassment probability	
	θ^r, θ^l		θ^r, θ^l		θ^r	
Constant	1.022	(0.124)	1.071	(0.240)		
Lender type: Harsh	0.027	(0.027)	-0.019	(0.018)	0.383	(1.820)
Lender type: Very harsh	0.103	(0.071)	0.013	(0.029)	1.552	(5.481)
Harassment probability	0.257	(0.032)	0.384	(0.203)		
Prevailing interest rate	-2.395	(0.182)	-2.208	(1.044)		
No lending history	-0.770	(0.046)	0.320	(0.418)	-0.085	(0.117)
Number of previous loans	0.024	(0.008)	0.002	(0.004)	-0.084	(0.131)
Number of previous loans squared	-0.003	(0.001)	-0.001	(0.000)	0.004	(0.007)
Number of missed payments in last loan	-0.087	(0.004)	-0.054	(0.045)	-0.190	(0.074)
Asked for loan under the influence of alcohol	-0.024	(0.015)	0.001	(0.008)	-0.448	(0.491)
Age	0.007	(0.002)	0.004	(0.003)	-0.047	(0.033)
Has post-secondary education	-0.070	(0.030)	-0.064	(0.021)	-0.939	(1.452)
Female	0.099	(0.040)	0.074	(0.070)	-0.569	(1.477)
Married (rel. to single)	0.018	(0.043)	0.025	(0.015)	-1.115	(1.152)
Divorced (rel. to single)	0.046	(0.044)	0.038	(0.061)	-0.211	(0.480)
Has children	-0.060	(0.042)	-0.039	(0.010)	0.006	(0.016)
Malaysian (rel. to Singaporean Chinese)	0.059	(0.033)	0.060	(0.053)	-0.009	(0.168)
Indian (rel. to Singaporean Chinese)	-0.023	(0.040)	0.006	(0.009)	-0.331	(0.359)
Currently a gang member	0.097	(0.036)	0.054	(0.115)	-0.947	(0.509)
Previously a gang member	0.037	(0.027)	0.045	(0.090)	-0.092	(0.225)
Number of previous convictions	-0.023	(0.012)	-0.005	(0.008)	-0.081	(0.126)
Drinks alcohol	-0.056	(0.052)	-0.071	(0.025)	-0.625	(0.273)
Uses drugs	-0.005	(0.030)	0.011	(0.005)	0.073	(0.102)
Frequents sex workers	0.057	(0.029)	0.011	(0.005)	0.289	(0.256)
Frequently treats friends	-0.142	(0.038)	-0.161	(0.350)	-0.268	(0.571)
Borrower's discounting, $\beta_i \times \delta_i$	0.174	(0.070)	0.094	(0.042)	-0.028	(0.060)
Borrower's risk aversion, γ_i	0.090	(0.061)	0.023	(0.023)	-0.454	(0.413)
Gamblers (in θ^r and θ^l)	0.538	(0.042)	0.409	(0.420)	1.871	(1.366)
Loan size					1.368	(1.161)
Low harshness level intercept					-5.832	(4.662)
Medium harshness level intercept					-2.287	(4.564)
High harshness level intercept					1.442	(1.919)
	Harassment cost θ^r		Harassment disutility θ^l		Effort cost θ^r	
Constant	0.437	(1.237)	1.302	(0.047)	1.528	(0.022)
Lender type: Harsh (multiplier for θ^r)	-0.221	(0.229)	0.248	(0.078)		
Lender type: Very harsh (multiplier for θ^r)	-0.119	(0.754)	1.616	(0.481)		
Post crackdown	1.050	(0.557)				

Robust standard errors clustered at the borrower level are shown in parentheses. For θ^r and θ^l , gang affiliation, prior convictions, addictions,

(c) Table 5: parameter estimates.

5.3 Table 7 and Figure 1: crackdown decomposition and cartel interest rates

Read:

- Table 7 shows that the crackdown lowers total lender profits from S\$3.08 million to S\$1.00 million, a 67.66% decline.

- Loan volume falls by 48.64%, while total harassment costs rise by 235.88%.
- Borrower surplus falls by 12.38%, missed payments increase by 30.78%, and the average number of times harassed increases by 31.02%.
- The decomposition shows that raising harassment costs alone would lower profits by 94.16%, while raising only the interest rate would increase profits by 36.16%.
- Figure 1 shows the cartel's profit response to alternative interest rates. Before the crackdown, profits are highest around 35%; after the crackdown, profitability is much lower and falls at high interest rates.

Economic message. The crackdown succeeds at reducing lender profits, but it does not mechanically improve borrower welfare. Lenders shrink loan sizes and borrowers face worse repayment incentives at higher rates.

Table 7

Decomposing the effects of the crackdown.

	No Crackdown (Level) (1)	Crackdown (Baseline) (Level) (2)	Overall (% Difference) (3)	Only $\kappa_{r,i}$ Increases (% Difference) (4)	Only r_i Increases (% Difference) (5)
Total lender profits (in \$m)	3.08	1.00	-67.66	-94.16	+ 36.16
Total loan volume (in \$m)	2.63	1.35	-48.64	-4.82	-46.44
Average harassment probability chosen	0.17	0.08	-51.10	-32.31	-26.82
Total interest revenue (in \$m)	6.99	6.66	-4.75	-4.36	-0.95
Total harassment costs (in \$m)	1.28	4.31	+ 235.88	+ 212.06	+ 3.03
Average borrower surplus (in \$0000)	0.51	0.45	-12.38	+ 2.06	-13.55
Average number of missed payments	4.58	6.00	+ 30.78	+ 1.04	+ 27.76
Average number of times harassed	1.93	2.52	+ 31.02	-6.78	+ 32.59

Column (3) shows the baseline (total) effects of the crackdown. Column (4) shows the effects of the crackdown if only the harassment cost, $\kappa_{r,i}$, increased. Column (5) shows the effects of only the nominal interest rate, r_i , increasing from 20% to 35%.

(a) Table 7: decomposing the crackdown.

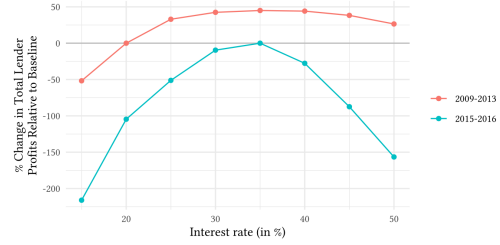


Fig. 1. Percentage change in total lender profits at alternative interest rates before and after the crackdown.

(b) Figure 1: lender profits at alternative rates.

5.4 Figure 2: targeting borrowers by repayment ability

Read:

- Borrowers are sorted into 20 equal loan-demand groups based on repayment ability.
- Removing the worst borrowers has little effect on lender profits because these borrowers are expensive to serve and require costly harassment.
- Removing middle-repayment-ability borrowers hurts lender profits most because they are large enough borrowers and still generate missed-payment revenue.
- Removing the highest repayment-ability borrowers lowers loan volume but has little effect on lender profits because these borrowers miss fewer payments and generate less penalty revenue.

Economic message. The most effective borrower-targeting policy is not to target the riskiest borrowers, but to target the profitable middle of the repayment distribution.

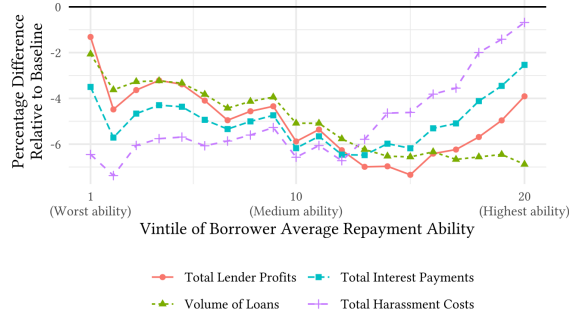


Fig. 2. Effects of targeting borrowers on lender outcomes.

Figure 2: effects of targeting borrowers on lender outcomes.

5.5 Table 8 and Figure 3: indirect interventions

Read:

- Table 8 shows that stopping a borrower from gambling reduces median lender profits by 26.16% and raises borrower surplus by 23.09%.
- Stopping drug use lowers profits by 13.62% and raises borrower surplus by 7.68%.
- Removing present bias lowers profits by 46.09%, while making borrowers both unbiased and more patient lowers profits by 57.07%.
- Figure 3 shows that discounting interventions have the largest effect for initially impatient borrowers, but still reduce lender profits for more patient borrowers.

Economic message. Reducing addictions and present-biased borrowing lowers loan demand and improves repayment effort. These interventions weaken the lender's business model from the demand side rather than through police enforcement alone.

Intervention	Median Impact on	
	Lender Profits	Borrower Surplus
Stop a borrower gambling	-26.16	23.09%
Stop a borrower using drugs	-13.62	7.68%
Remove a borrower's present bias (setting borrower's β_i to 1)	-46.09	6.66%
Set a borrower's β_i to 1 and δ_i to 0.999 (0.95 annual discount factor)	-57.07	13.45%

(a) Table 8: indirect interventions.

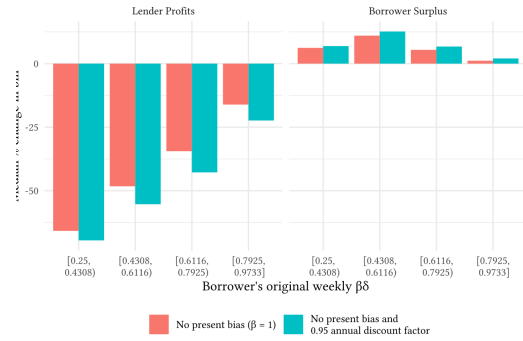


Fig. 3. Heterogeneous effects of discounting interventions.

(b) Figure 3: discounting interventions.

6 Short summary

- The paper studies illegal money lending using a unique panel of Singapore loan-shark borrowers.
- The full data cover 1,090 borrowers and 11,032 loans over 2009–2016.
- The market is cartelized at the contract-term level but decentralized at the lender level for loan-size and harassment choices.
- Borrowers request loans and choose lenders; lenders decide whether to lend, how much to lend, and how harsh to be.
- The model includes borrower present bias, discounting, risk aversion, effort, harassment disutility, lender learning, and lender harassment costs.
- The enforcement crackdown raises harassment costs and induces a cartel interest-rate increase from 20% to 35%.
- Lender profits fall by 67.7% and loan volume falls by 48.6% after the crackdown.
- Borrower welfare falls by 12.4%, because borrowers receive smaller loans and face worse repayment incentives.
- The cost-channel and rate-channel point in different directions: higher enforcement costs hurt lenders, but higher rates can partly raise profits.
- Borrower targeting is most effective when aimed at the middle of the repayment-ability distribution.
- Reducing gambling, drug use, present bias, or heavy discounting reduces lender profits and improves borrower surplus.
- The main policy message is that law enforcement can suppress illegal lenders, but borrower-side interventions can also weaken the market by reducing demand and improving repayment behavior.

7 Conclusion

7.1 Core conclusion

Leong, Li, Pavanini, and Walsh estimate a structural model of an illegal lending market and use it to quantify the effects of policy interventions. The enforcement crackdown lowers lender profitability substantially, but it also reduces borrower welfare because borrowers face smaller loans, higher interest rates, more missed payments, and more harassment events.

7.2 Economic intuition

The illegal lending market survives because it combines high borrower demand, present-biased repayment behavior, loan-reset penalties, and coercive enforcement. Lenders make profits not simply by charging high rates, but by exploiting missed payments and enforcing repayment through harassment. Increasing enforcement pressure raises lenders' cost of harassment and shrinks the market, but borrower-side policies can be especially powerful because they reduce loan demand and improve repayment incentives.

7.3 Takeaway

The paper shows how structural estimation can be used to evaluate policy in a market where direct experimentation is infeasible. The main lesson is that eliminating illegal money lending requires both supply-side enforcement against lenders and demand-side interventions that reduce the behavioral and addiction-driven forces pushing borrowers into loan-shark credit.